

George Gu

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EDUCATION

University of Michigan

Bachelor of Science in Engineering, Computer Science

May 2028

Ann Arbor, MI

- **GPA:** 3.81/4.00
- **Coursework:** Machine Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Theory of Computation, Computer Organization, Web Systems, Computer Security

EXPERIENCE

Capital One

Technology Intern

June – August 2026

Richmond, VA

- Architected data pipelines via scheduled **AWS Lambda** workers to ingest daily snapshots from internal asset-management systems, yielding clean tabular datasets to train predictive models on application lifecycle and business criticality.
- Engineered Python **REST APIs** orchestrating data-attestation and compliance workflows, securing them with parameterized PostgreSQL queries and role-based access control across corporate domains and org hierarchies.
- Authored **20+** automated integration tests covering mock database transactions, cross-region failures, and network-timeout edge cases, lifting repository-wide coverage to **91%** for the compliance service.

BoilerVault

Founding Engineer (Contract)

January 2026 – Present

West Lafayette, IN

- Built a multi-tenant operations platform with FastAPI, PostgreSQL, and Next.js/TypeScript for a **3-campus** student storage business, implementing JWT auth and 20+ REST endpoints validated by **140+** tests.
- Reconciled out-of-order payment events across **4 Stripe accounts** and **3 WordPress booking sites** via idempotent webhook pipelines, automating customer, order, and billing creation with 0 manual entry across **\$130K+** in gross revenue.
- Migrated **2,000+** legacy records via three-tier matching (exact email, rapidfuzz fuzzy name, Stripe charge fallback), replacing a failing Zapier workflow and deploying on Railway/Vercel as the production system of record.

Nexteer Automotive

Software Engineering Intern

May – August 2025

Saginaw, MI

- Developed an **LLM-powered** IDE extension in Python and TypeScript that parsed **300+** internal engineering guidelines and automated compliance checks across C and H files, integrating Azure AI and Copilot APIs.
- Designed prompt pipelines and few-shot strategies to extract rules from unstructured documentation, tuning the system to **95% violation-detection accuracy** and saving engineers an estimated 8 hours of review per week.
- Deployed the tool across **26 sites** and integrated it with internal dev pipelines, cutting manual code-review time by **87.5%** for safety-critical embedded steering systems.

PROJECTS

Image Classification & Transfer Learning

Machine Learning Developer

January – April 2026

Ann Arbor, MI

- Implemented a **Vision Transformer** from scratch in **PyTorch**, building patch-to-token projection, a learnable classification token, sinusoidal positional embeddings, and multi-head self-attention encoder blocks.
- Engineered a 2-stage pretrain/fine-tune pipeline, sweeping augmentation intensity across 4 levels and tuning early-stopping patience on validation AUROC to correct overfitting, reaching **0.93 accuracy** and **0.97 AUROC** on validation.

Scalable Search Engine

Systems Developer

January – April 2026

Ann Arbor, MI

- Built a multi-stage MapReduce pipeline in Python to construct an inverted index over **3,000+ documents**, computing TF-IDF scores and document-normalization factors across a parallel, multi-job architecture.
- Implemented a REST API index server in Flask serving **3 partitioned inverted index segments**, integrating PageRank-weighted cosine similarity scoring to balance TF-IDF with link authority when ranking results.

Resume Screener

Machine Learning Developer

January – May 2025

Ann Arbor, MI

- Built an **NLP** pipeline over a labeled dataset of **1,000+ resumes and job postings**, applying tokenization, TF-IDF vectorization, and feature engineering for resume-to-posting matching.
- Designed **scikit-learn** and **TensorFlow** models to rank resumes against job postings, using sentence-transformer embeddings and **RAG** retrieval over job descriptions, deployed via Streamlit to **100+ users**.

SKILLS/ADDITIONAL

Languages: Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML, CSS

Frameworks/Developer Tools: PyTorch, TensorFlow, Git, Docker, Kubernetes, Unix, Node.js, React.js, Flask, PostgreSQL

Interests: Philosophy and Metaphysics, Game Design, Boulderling, Skateboarding, Golf, Guitar, Calisthenics

PUBLICATION

Gu G, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, “A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle”, 2023, *Antibiotics*, 12, 1509.

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